# PSTAT 231 HW 1

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**Question 1**:

Define supervised and unsupervised learning. What are the difference(s) between them?

Supervised learning fits a model where the target is known and the goal is to learn the mapping from predictor X to response Y to be able to predict Y for new data. Unsupervised learning uses predictors to find structure in the data. The main difference between the 2 is that in supervised learning, we know the target outcome while unsupervised learning trues to find patterns and structure based on the data.

**Question 2**:

Explain the difference between a regression model and a classification model, specifically in the context of machine learning.

A regression model predicts a quantitative response variable while a classification model predicts a qualitative response variable.

**Question 3**:

Name two commonly used metrics for regression ML problems. Name two commonly used metrics for classification ML problems

Regression - MSE and $R^2$ Classification - Accuracy and ROC AUC

**Question 4**:

As discussed, statistical models can be used for different purposes. These purposes can generally be classified into the following three categories. Provide a brief description of each.

Descriptive models: Summarize or visualize the characteristics of data to show trends or patterns.

Inferential models: Used to test hyptheses or estimate relationships between variables.

Predictive models: Used to predict responses for new data.

**Question 5**:

Define mechanistic. Define empirically-driven. How do these model types differ? How are they similar?

A mechanistic model assumes a specific function form for f while an empirically driven model makes no strong asssumptions about the form of f and lets the data determine the shape.

In general, is a mechanistic or empirically-driven model easier to understand? Explain your choice.

A mechanistic model is easier to understand in my opinion since they have a fixed functional form with a small number of parameters. Empirically-driven models tend to produce complex fits that are hard to translate into a simple relationship between predictors and response.

Describe how the bias-variance tradeoff is related to the use of mechanistic or empirically-driven models.

Mechanistic modles are less flexible so they tend to have higher bias but lower variance while empirically driven models tends to have the opposite (low bias, high variance).

**Question 6**:

A political candidate's campaign has collected some detailed voter history data from their constituents. The campaign is interested in two questions:

Given a voter's profile/data, how likely is it that they will vote in favor of the candidate?

How would a voter's likelihood of support for the candidate change if they had personal contact with the candidate?

Classify each question as either predictive or inferential. Explain your reasoning for each.

The first question is predictive because we want the likelihood of support given a voter's profile. The goal is an accurate prediction of support, not understanding why they chose to support the party

The 2nd question is inferential since it is asking about the relationship between having personal contact and the likelihood of support. Essentially, it asks how the response changes based on a predictor variable change.

**Exploratory Data Analysis**

This section will ask you to complete several exercises. For this homework assignment, we'll be working with the mpg data set that is loaded when you load the tidyverse. Make sure you load the tidyverse and any other packages you need.

Exploratory data analysis (or EDA) is not based on a specific set of rules or formulas. It is more of a state of curiosity about data. It's an iterative process of:

generating questions about data visualize and transform your data as necessary to get answers use what you learned to generate more questions

A couple questions are always useful when you start out. These are "what variation occurs within the variables," and "what covariation occurs between the variables."

You should use the tidyverse and ggplot2 for these exercises.

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.2.1     v readr     2.2.0
## v forcats   1.0.1     v stringr   1.5.2
## v ggplot2   4.0.0     v tibble    3.3.1
## v lubridate 1.9.5     v tidyr     1.3.2
## v purrr     1.2.2
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

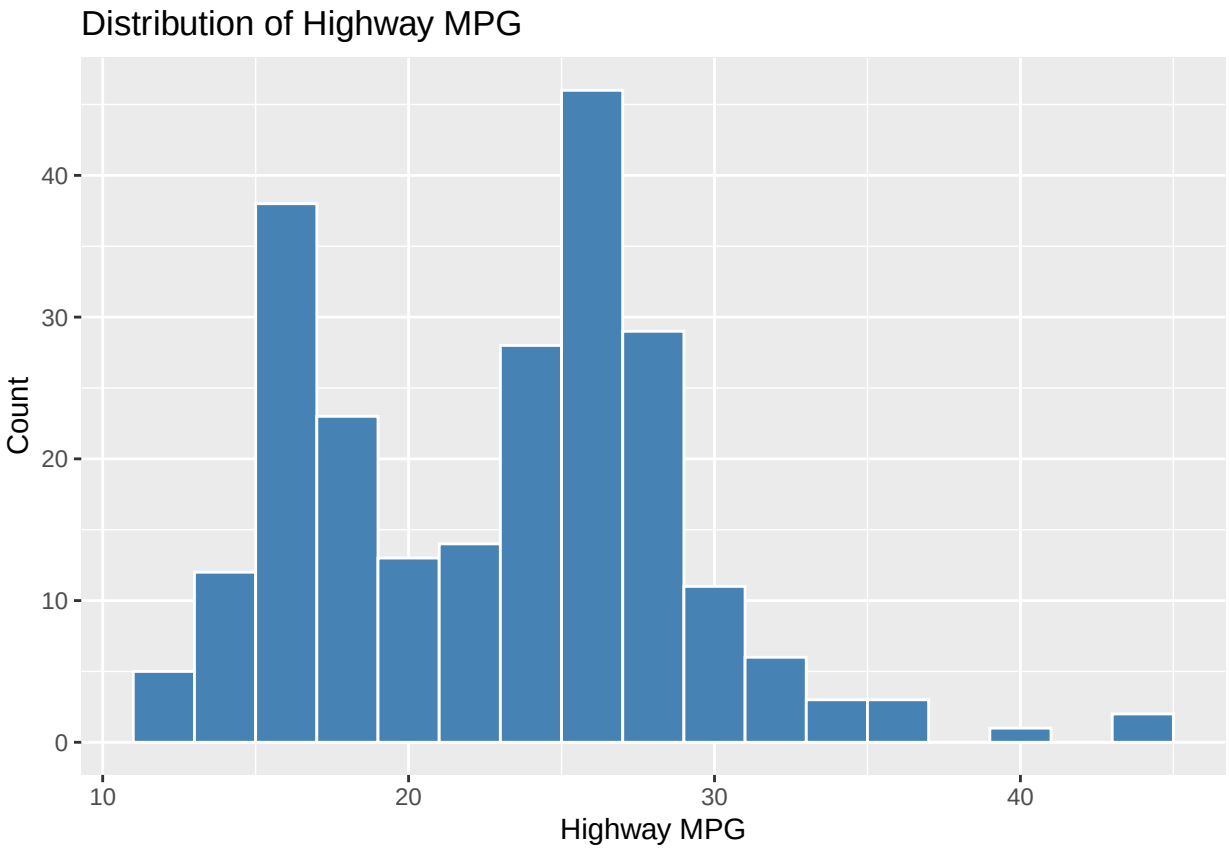
```
library(corrplot)
```

```
## corrplot 0.95 loaded
```

**Exercise 1**:

We are interested in highway miles per gallon, or the hwy variable. Create a histogram of this variable. Describe what you see/learn.

```
ggplot(mpg, aes(x = hwy)) +
  geom_histogram(binwidth = 2, fill = "steelblue", color = "white") +
  labs(title = "Distribution of Highway MPG", x = "Highway MPG", y = "Count")
```

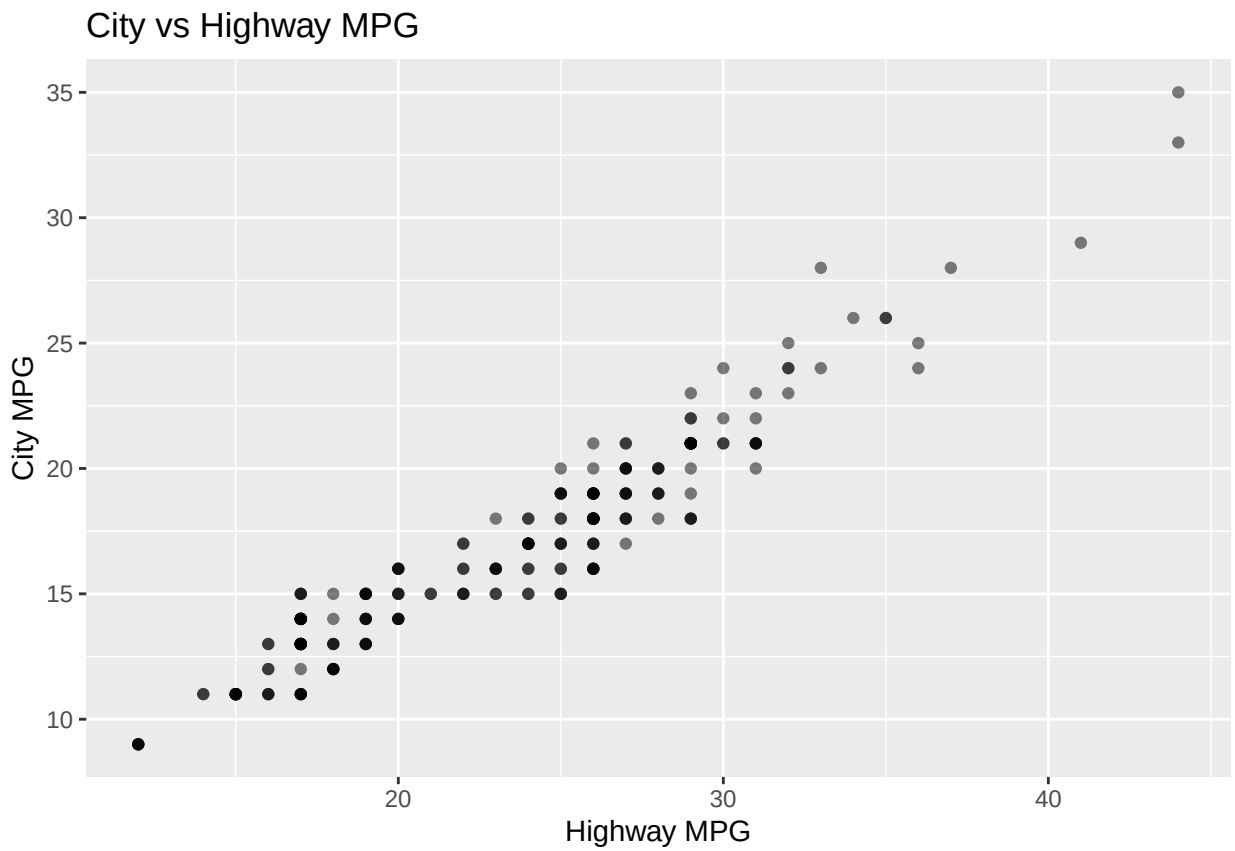



Highway MPG ranges from about 12 to 44, with most cars between 15 and 30 MPG. I notice that the distribution is bimodal with one peak at the 16-18 mark and another at the 26-28 mark. The two groups most likely represent different vehicle types.

**Exercise 2**:

Create a scatterplot. Put hwy on the x-axis and cty on the y-axis. Describe what you notice. Is there a relationship between hwy and cty? What does this mean?

```
ggplot(mpg, aes(x = hwy, y = cty)) +
  geom_point(alpha = 0.5) +
  labs(title = "City vs Highway MPG", x = "Highway MPG", y = "City MPG")
```

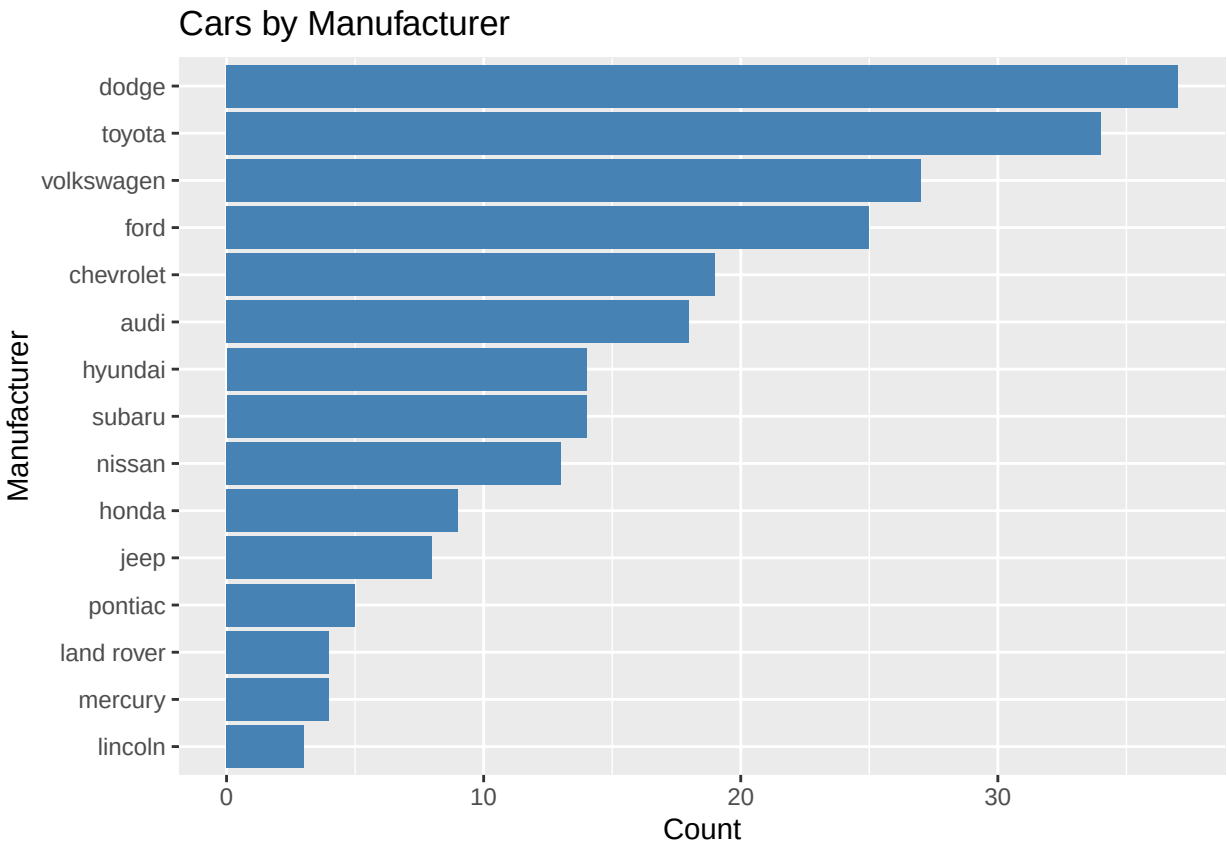



There is a very strong positive linear relationship between highway MPG and city MPG. This means that cars with high hwy mpg also have high cty mpg. This also means that the two variables carry largely the same information about fuel efficiency so if you were to use both predictors in a model, they would be highly collinear.

**Exercise 3**:

Make a bar plot of manufacturer. Flip it so that the manufacturers are on the y-axis. Order the bars by height. Which manufacturer produced the most cars? Which produced the least?

```
ggplot(mpg, aes(y = fct_infreq(manufacturer))) +
  geom_bar(fill = "steelblue") +
  scale_y_discrete(limits = rev) +
  labs(title = "Cars by Manufacturer", x = "Count", y = "Manufacturer")
```



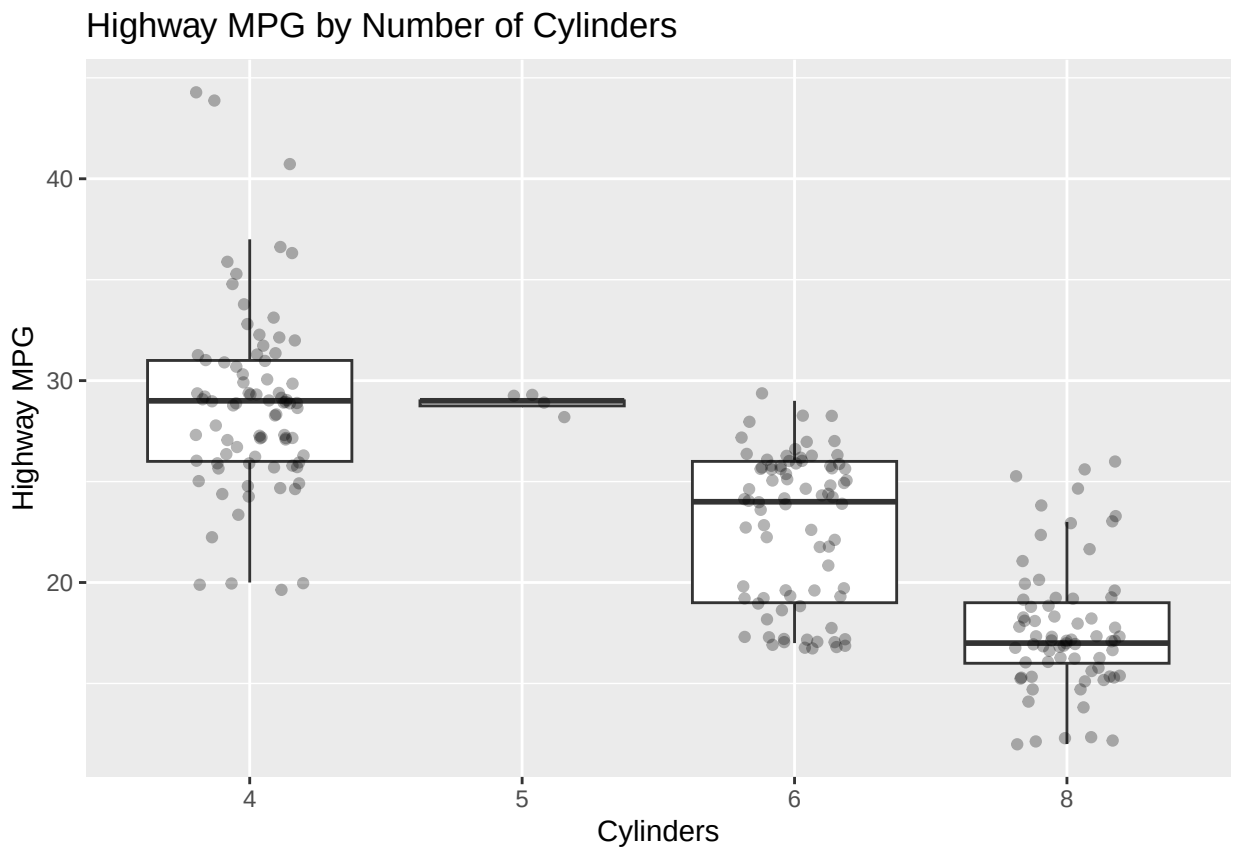

Dodge produced the most cars while Lincoln produces the fewest.

**Exercise 4**:

Make a box plot of hwy, grouped by cyl. Use geom_jitter() and the alpha argument to add points to the plot.

Describe what you see. Is there a relationship between hwy and cyl? What do you notice?

```
ggplot(mpg, aes(x = factor(cyl), y = hwy)) +
  geom_boxplot(outlier.shape = NA) +
  geom_jitter(width = 0.2, alpha = 0.3) +
  labs(title = "Highway MPG by Number of Cylinders", x = "Cylinders", y = "Highway MPG")
```



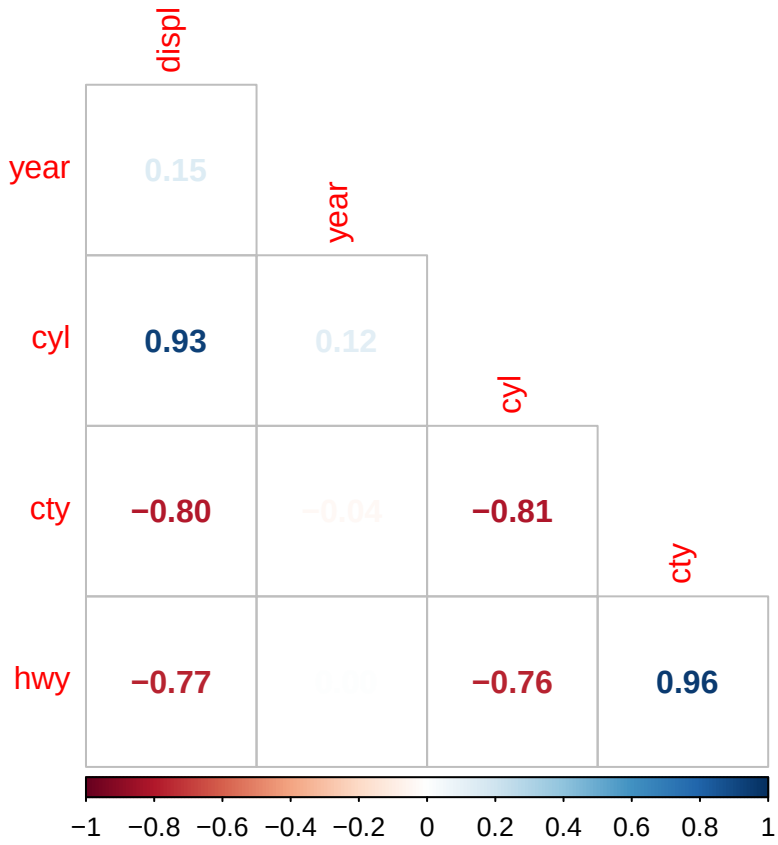

There is a clear negative relationship. Highway MPG decreases as the number of cylinders increases. Median MPG is highest for 4-cylinder cars and lowest for 8-cylinder cars. This makes sense, because more cylinders usually means a larger engine that burns more fuel.

**Exercise 5**:

Use the corrplot package to make a lower triangle correlation matrix of the mpg dataset. (Hint: You can find information on the package here.)

Which variables are positively or negatively correlated with which others? Do these relationships make sense to you? Are there any that surprise you?

```
mpg %>%
  select(where(is.numeric)) %>%
  cor() %>%
  corrplot(method = "number", type = "lower", diag = FALSE)
```



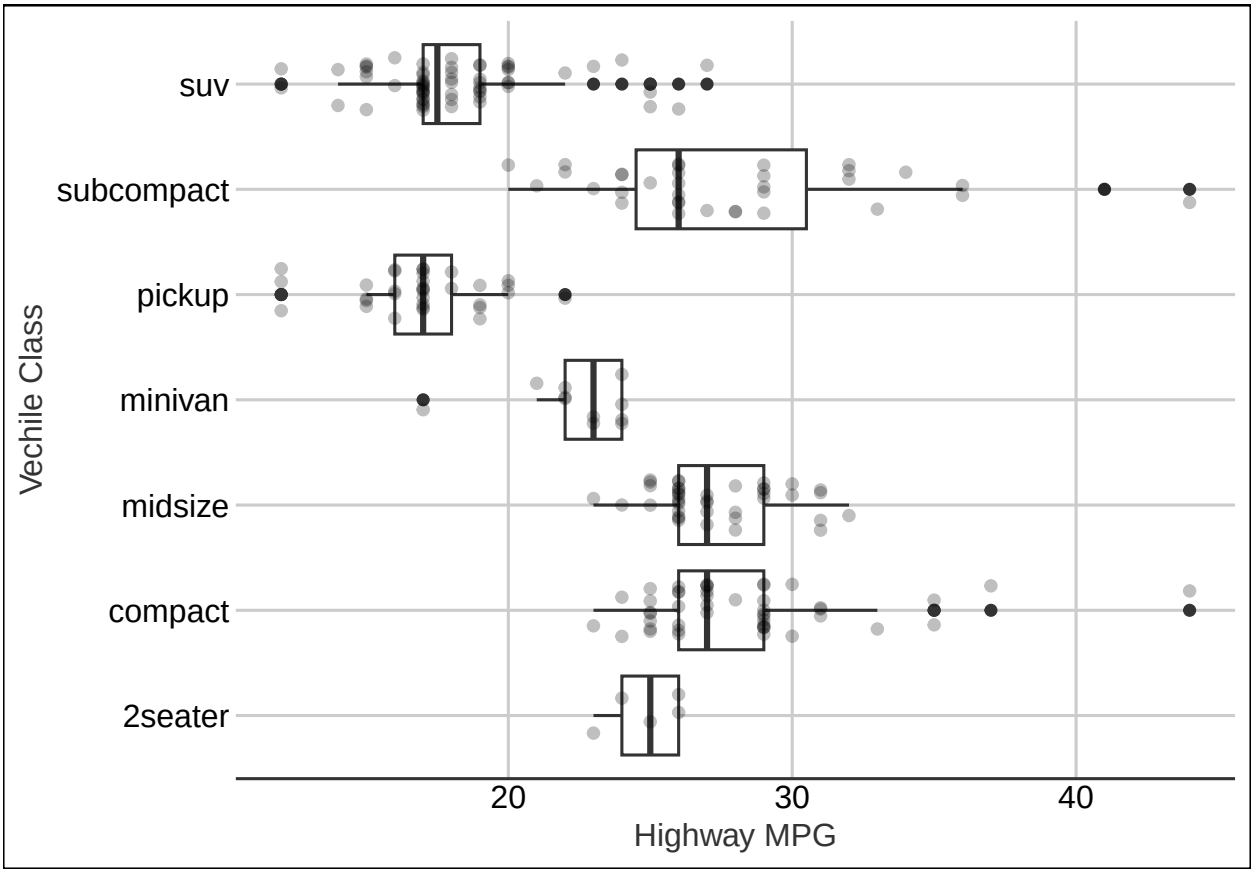

cty and hwy as well as displ and cyl are positively correlated while displ and cyl with cty and hwy are both negatively correlated while year is near zero with every other variable. Most of these relationships make sense since as I mentioned earlier larger engines tend to have more cylinders and therefore have lower fuel efficiency.

Exercise 6:

USing ggplot(), recreate the following graphic as closely as you can. Hint: Use the ggthemes package.

```
library(ggthemes)

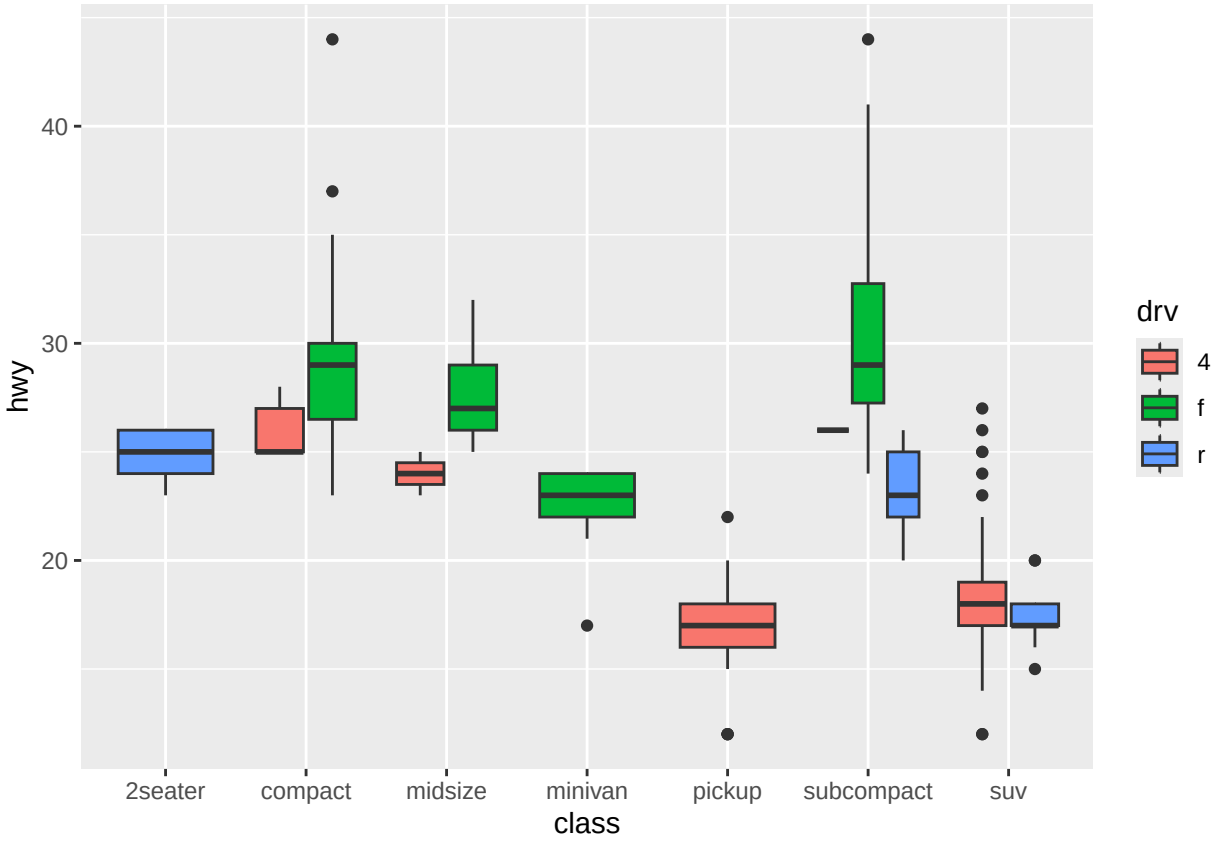
ggplot(mpg, aes(x = hwy, y = class)) +
  geom_boxplot() +
  geom_jitter(alpha = 0.25, width = 0, height = 0.25) +
  theme_gdocs() +
  labs(x = "Highway MPG", y = "Vechile Class")
```




Exercise 7

Recreate the following graphic.

```
ggplot(mpg, aes(x = class, y = hwy, fill = drv)) + geom_boxplot()
```

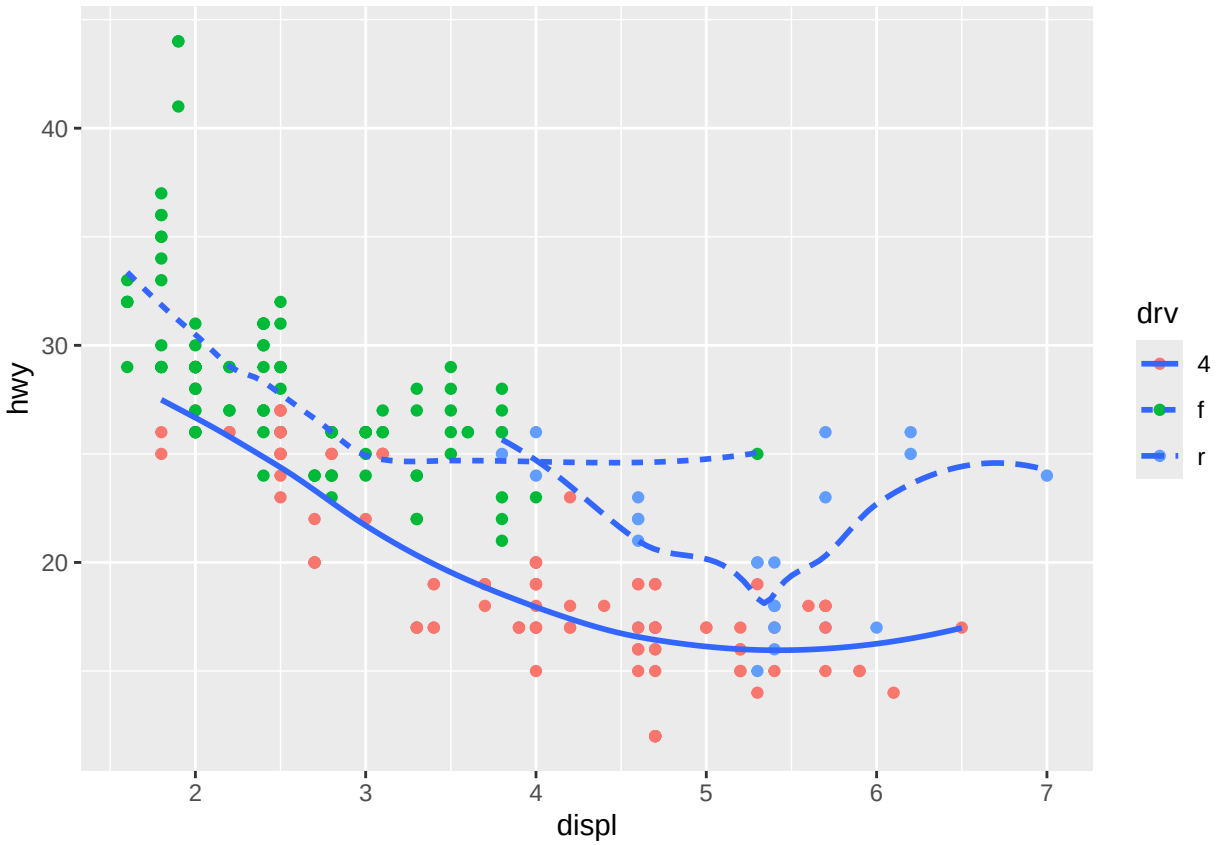



Exercise 8

Recreate the following graphic.

```
ggplot(mpg, aes(x = displ, y = hwy)) + geom_point(aes(color = drv)) + geom_smooth(aes(linetype = drv), s
```

```
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```


hwy
40
30
20
2
3
4
5
6
7
displ
drv
4
f
r